cp_7b

1.

Print Length Times | 30/30

Solved

Easy

Write a program that reads a string and prints the first character of the given string on `N` lines, where `N` is the length of the given string.

Input

The input will be a single line containing a string.

Output

The output should be `N` lines, with each line containing a string that is the first character of the given string. Here, `N` is the length of the given string.

Explanation

For example, if the given string is **Cool**,

- The first character in the string is **C**.
- The length of the given string is **4**
- **C** should be printed **4** times.

The output should be,

```
C
C
C
C
```

Sample Input 1

```
Cool
```

Sample Output 1

```
C
C
C
C
```

Sample Input 2

```
Program
```

Sample Output 2

```
P
P
P
P
P
P
P
```

code:

```
n=input()
n_len=len(n)
count=0
while count<n_len:
    print(n[0])
    count=count+1
```

```
2
```

Print the Characters | 35/35

Solved

Easy

Write a program that reads a string and prints each character of the given string on a new line.

Input

The input will be a single line containing a string.

Output

The output should be `N` lines, with each line containing a string that is one of the characters of the given string. Here, `N` is the length of the given string.

Explanation

For example, if the given string is **shine**,

- The length of the given string is **5**.
- Each character of the string should be printed on a new line.

The output should be,

```
s
h
i
n
e
```

Sample Input 1

```
shine
```

Sample Output 1

```
s
h
i
n
e
```

Sample Input 2

```
Morning
```

Sample Output 2

```
M
o
r
n
i
n
g
```

code:

```python
n=input()
n_len=len(n)
count=0
while count<n_len:
    print(n[count])
    count=count+1
```

3

Square - 2 | 25/25

Easy

Write a program that reads a number `N` and prints a **Square** of `N` rows and `N` columns using numbers starting from **1**.

```
111
222
333
```

Input

The input will be a single line containing an integer representing `N` .

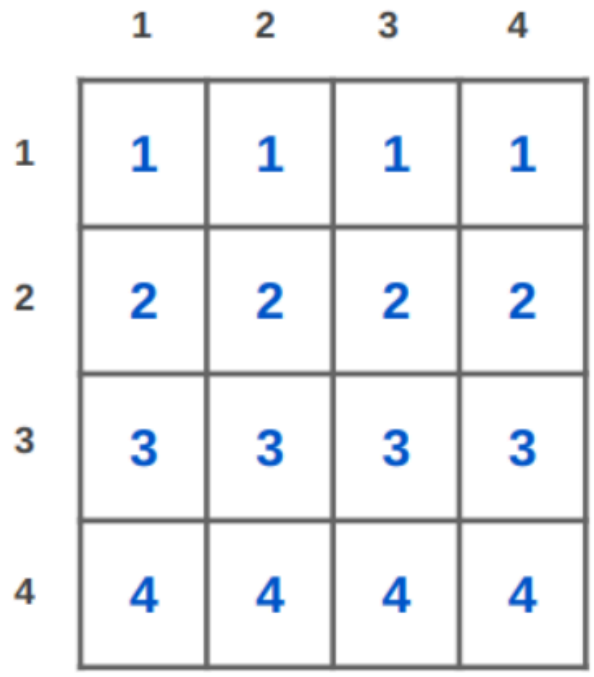
Output

The output should be `N` rows containing numbers as a **Square** shown in the sample output.

Explanation

For example, if the given number is `N = 4` ,

- Print the numbers on **4** rows and **4** columns as shown in the given image.



The output should be,

```
1111
```

code:

```
num=int(input())
count=1
while count<=num:
    str_count=str(count)
    print(str_count*num)
    count=count+1
```

4

Rectangle | 35/35

Easy

Write a program that reads two numbers `M` and `N`, and prints a **Rectangle** of `M` rows and `N` columns using stars (`*`).

```
*****
*****
*****
*****
```

Input

The first line of input contains an integer representing `M`.
The second line of input contains an integer representing `N`.

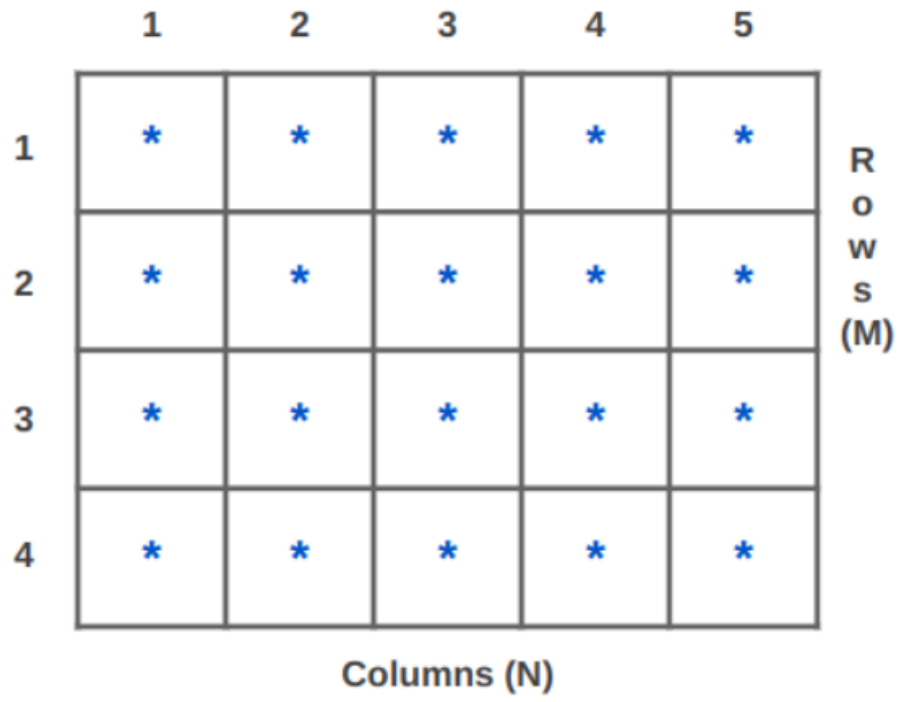
Output

The output should be `M` rows and `N` columns containing stars (`*`) as a **Rectangle** shown in the sample output.

Explanation

For example, if the given numbers are `M = 4` and `N = 5`,

- Print the stars (`*`) on **4** rows and **5** columns as shown in the given image.



code:

```
m=int(input())
n=int(input())
count=0
while count<m:
```

```
        print("*"*n)
        count=count+1
```

5

Right Angled Triangle | 40/40

Easy

Write a program that reads a number `N` and prints a **Right Angled Triangle** of `N` rows using stars (`*`).

```
*
**
***
****
```

Input

The input will be a single line containing an integer representing `N` .

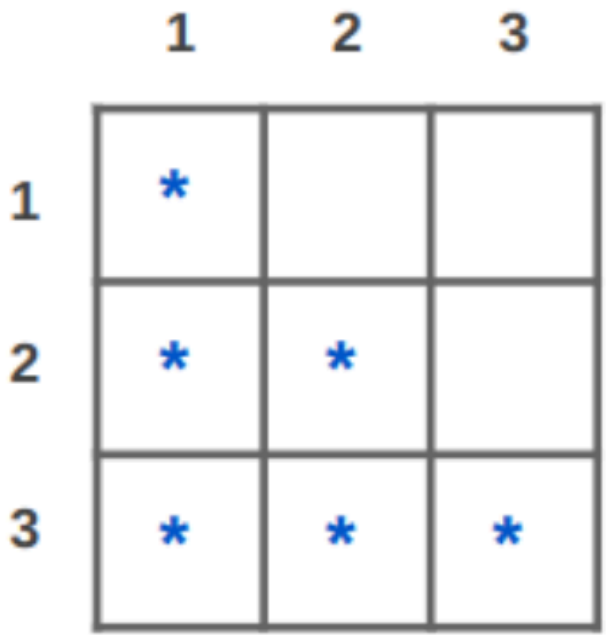
Output

The output should be `N` rows containing stars (`*`) as a **Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 3` ,

- Print the stars (`*`) on **3** rows as shown in the given image.



code:

```
n=int(input())
count=1
while count<=n:
    print("*"*count)
    count=count+1
```

6

Right Angled Triangle - 2 | 40/40

Easy

Write a program that reads a number `N` and prints a **Right Angled Triangle** of `N` rows using numbers starting from **1**.

```
1
2 2
3 3 3
4 4 4 4
```

Note

There is a space after every number.

Input

The input will be a single line containing an integer representing `N` .

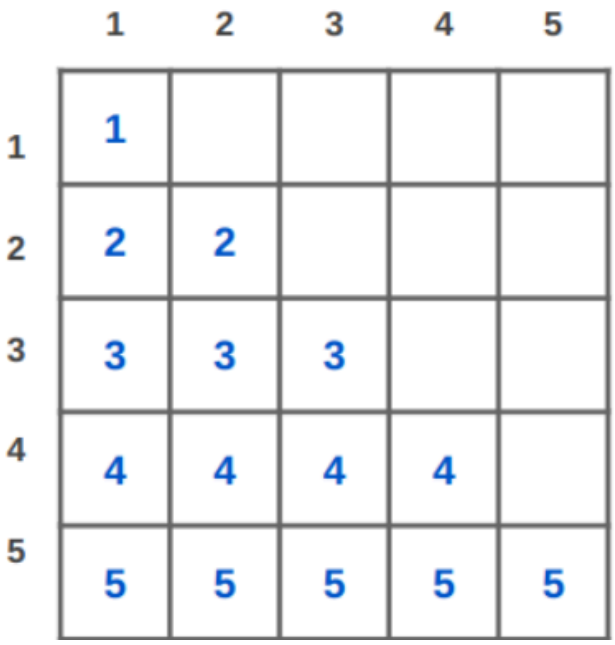
Output

The output should be `N` rows containing numbers as a **Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 5` ,

- Print the numbers on **5** rows as shown in the given image.



code:

```
n=int(input())
count=1
while count<=n:
```

```
    str_count=str(count)
    print((str_count+" ")*count)
    count=count+1
```

7

Right Angled Triangle - 3 | 40/40

Easy

Write a program that reads a number `N` and prints a **Right Angled Triangle** of `N` rows using stars (`*`) and pluses (`+`).
The first `N - 1` rows contains stars (`*`) and the `Nth` row contains pluses (`+`).

```
*
* *
+ + +
```

Note

There is a space after every star (`*`) and plus (`+`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) and pluses (`+`) as a **Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 4` ,

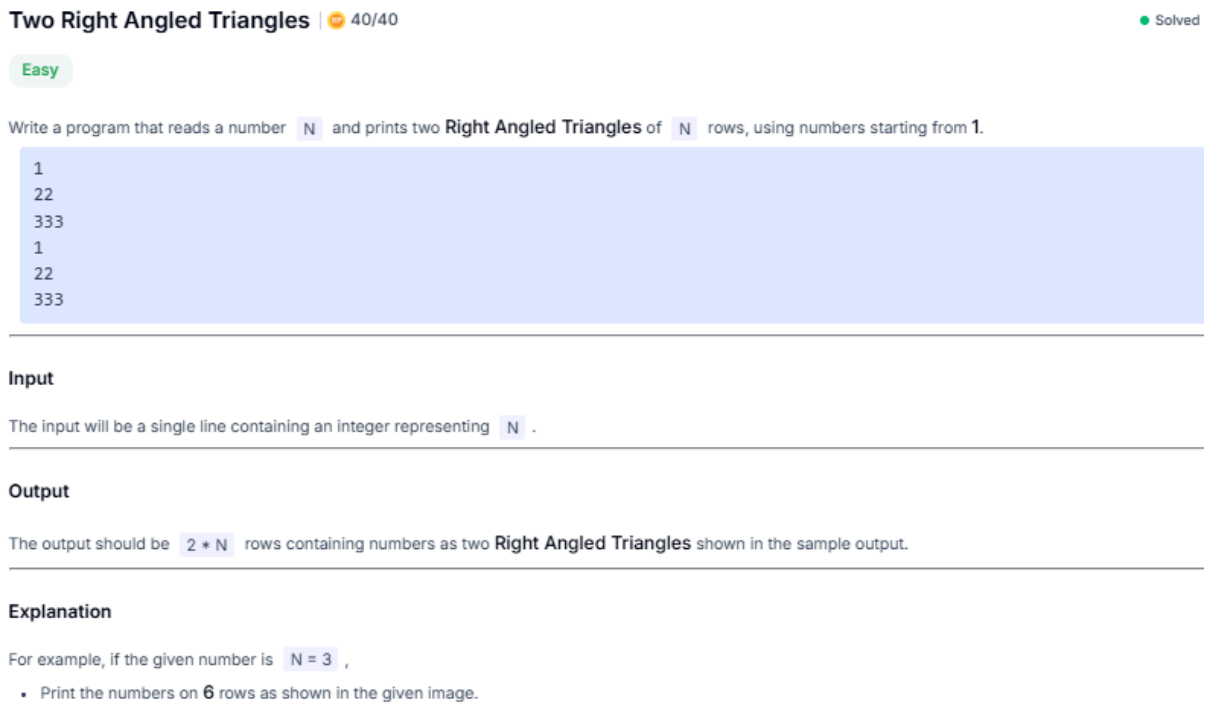
- Print stars (`*`) and pluses (`+`) on **4** rows as shown in the given image.
  - The first **3** rows should contain stars (`*`).
  - The **4th** row should contain `N` pluses (`+`).

code:

```
n=int(input())
count=1
while count<=n-1:
    print("* "*count)
    count=count+1
if count==(n):
```

```
    print("+ "*count)
```

8



Two Right Angled Triangles | 40/40 ● Solved

Easy

Write a program that reads a number `N` and prints two **Right Angled Triangles** of `N` rows, using numbers starting from **1**.

```
1
22
333
1
22
333
```

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `2 * N` rows containing numbers as two **Right Angled Triangles** shown in the sample output.

Explanation

For example, if the given number is `N = 3` ,

- Print the numbers on **6** rows as shown in the given image.

code:

```
n=int(input())
count=1
while count<=n:
    str_count=str(count)
    print(str_count*count)
    count=count+1
count=1
while count<=n:
    str_count=str(count)
    print(str_count*count)
    count=count+1
```

9

Square | 30/30

Write a program that reads a number `N` and prints a **Square** of `N` rows and `N` columns using stars (`*`).

```
* * *
* * *
* * *
```

Note

There is a space after every star(`*`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as a **Square** shown in the sample output.

Explanation

For example, if the given number is `N = 4` ,

- Print the stars (`*`) on **4** rows as shown in the given image.

code:

```
n=int(input())
count=0
while count<n:
    print("* "*n)
    count=count+1
```

10

Rectangle - 2 | 30/30

Solv

Easy

Write a program that reads two numbers `M` and `N`, and prints a **Rectangle** of `M` rows and `N` columns using pluses (`+`).

```
+ + + + + +
+ + + + + +
+ + + + + +
+ + + + + +
```

Note

There is a space after every plus (`+`).

Input

The first line of input contains an integer representing `M`.
The second line of input contains an integer representing `N`.

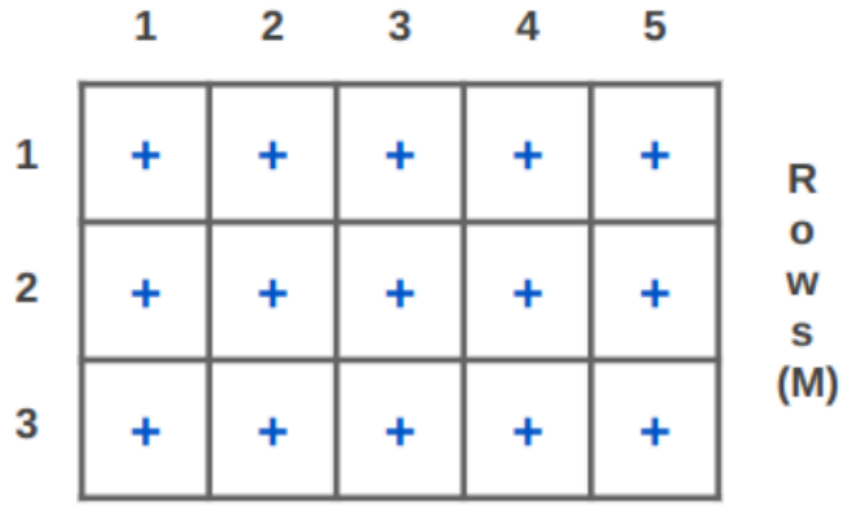
Output

The output should be `M` rows and `N` columns containing pluses (`+`) as a **Rectangle** shown in the sample output.

Explanation

For example, if the given numbers are `M = 3` and `N = 5`,

- Print the pluses (`+`) on **3** rows and **5** columns as shown in the given image.



code:

```
m=int(input())
n=int(input())
count=0
while count<m:
```

```
    print("+ "*n)
    count=count+1
```